Soc Internship day 20

1. Explain Forensics

Digital Forensics is the process of identifying, collecting, preserving, analyzing, and presenting digital evidence from computers, mobile devices, networks, or other electronic systems in a way that is legally acceptable.

Purpose:

- Investigate cybercrimes
- Identify attackers
- Recover deleted data
- Determine how an incident occurred
- Support legal proceedings

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2. List Types of Digital Evidence

- **System logs** (Windows Event Logs, Linux logs)
- **Network logs** (firewall, IDS/IPS, proxy logs)
- **Hard disk data** (documents, images, deleted files)
- **Memory (RAM) dumps**
- **Email messages**
- **Browser history and cookies**
- **Chat and messaging records**
- **USB connection history**
- **Registry files (Windows)**
- **File metadata** (creation, modification, access times)
- **Mobile device data** (calls, SMS, apps, GPS)
- **Cloud storage records**

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3. Describe the Digital Forensics Steps

1. **Identification**
    - Detect the incident and identify potential sources of evidence.
2. **Preservation**
    - Secure the evidence and prevent any modification.

  - Maintain the **chain of custody** to document who handled the evidence.
3. **Collection**
   - Acquire forensic copies (bit-by-bit images) of storage devices and collect volatile data like RAM when necessary.
4. **Examination**
   - Recover deleted files, extract artifacts, and organize the collected data for analysis.
5. **Analysis**
   - Correlate evidence to determine what happened, when it happened, how it happened, and who was involved.
6. **Documentation and Reporting**
   - Prepare a detailed report describing the methods used, evidence collected, findings, and conclusions in a clear and legally defensible manner.

Example

A company suspects unauthorized access to a server:

- **Identification:** Detect suspicious login activity.
- **Preservation:** Isolate the server and preserve its state.
- **Collection:** Create a forensic disk image and capture memory.
- **Examination:** Review logs and recover deleted files.
- **Analysis:** Find that an attacker used stolen credentials to access sensitive data.
- **Reporting:** Document the investigation and provide evidence for incident response or legal action.

This workflow ensures that digital evidence remains reliable and admissible throughout an investigation.